Super Magnifier LV

Update Manual · Version 2026.09.08.0002

Recent release notes. See CHANGELOG.md in the install folder for full history.

[2026.09.08.0002] -- 2026-09-08

Fixes from `SuperMagnifierLV_Full_Review_2026-09-07.md`'s Critical/High findings. Source-level changes only in the session that made them (no `dotnet` toolchain available) -- pending a local build confirmation.

Fixed

- **Stale virtual-screen-bounds cache on monitor topology change while the lens stayed put.** The cache was only invalidated when the lens itself crossed onto a differently-named monitor (a once-a-second poll) -- unplugging or reconfiguring a display elsewhere could leave the lens clamped to now-nonexistent screen area, or unable to reach newly added area, until an unrelated lens move happened to trigger the poll. `MainWindow` now subscribes directly to `Microsoft.Win32.SystemEvents.DisplaySettingsChanged` and invalidates the cache immediately on any display-topology change.
- `CursorHighlighter`**'s overlay window was sized once at construction and never resized.** Adding a monitor after the highlighter started left the cursor ring unable to render on the new screen area. Now resizes from fresh `SystemParameters.VirtualScreen*` values on the same `DisplaySettingsChanged` event, and drops its per-monitor DPI cache at the same time since monitor handles can be reused for a different physical display after a topology change.
- **Crash-on-launch from a hand-edited or corrupted** `ThemeName` **in settings.** A typo'd or removed theme key built an unresolvable `Uri` and threw during window construction, before any UI was visible -- the worst possible failure mode for a low-vision accessibility tool. The new theme dictionary is now built and validated before the old one is touched, with a hard fallback if validation fails.
- `CursorHighlighter.GetScaleForPoint` **called** `MonitorFromPoint`**/**`GetDpiForMonitor` **uncached on every 16ms tick** (~2 Win32 calls/tick for the lifetime of an enabled highlighter). Now caches DPI per monitor handle, refreshed only on `DisplaySettingsChanged`.
- **Snap-to-word polling leaked a** `CancellationTokenSource` **per poll** -- cancelled on the next iteration but never disposed, a small but real resource leak over long reading sessions. Now disposed before each new one is created.

Changed

- Preset slot buttons (`Preset1Btn`...`Preset5Btn`) grown to `MinWidth="44"` in addition to the existing `MinHeight="44"` -- the rendered click target was under the WCAG AAA 44px floor width-wise despite meeting it height-wise.

Already fixed before this review (confirmed still good, not new this cycle)

- Dead `MainWindowViewModel` stub deleted; `DataContext = this` is the sole data-context assignment.
- Hardcoded `FocusRingBrush="#FFD700"` removed from `App.xaml` -- the ring now correctly resolves per-theme.
- `MagnifierEngine`'s font-smoothing calls use `SPIF_UPDATEINIFILE | SPIF_SENDCHANGE`, closing a persistent-corruption-on-crash risk.
- `ListViewItem`/`GridViewColumnHeader` styling added to `AppPresetDialog` -- the near-black-on-black dark-theme bug is fixed.

Known open items (not addressed this cycle)

- `MainWindow.xaml.cs` remains a ~2,936-line God Object with zero MVVM -- the largest maintainability risk in the app.
- `LensWindow`'s hover-bar `ContextMenu`/`MenuItem` is still unstyled, rendering OS-default chrome against dark themes.
- `AppPresetDialog`'s exe-name field still has no `AutomationProperties.Name`.
- `ResizeMode` is inconsistent between `MainWindow` (`CanResize`) and `AboutDialog` (`CanResizeWithGrip`).
- No root `README.md`/`CHANGELOG.md` existed before this pass (see note below).

Builds prior to 2026-09-08 were not individually changelogged; `version.txt`/`installer.nsi`/csproj were kept in sync throughout (confirmed no digit-width drift as of the 2026-09-01.0002 baseline). This CHANGELOG and a root `README.md` (promoted from `Docs/README.md`) are new as of this pass, per the review's own recommendation to add them for contributor/AI-agent onboarding parity with sibling apps.